

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5112

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 10%

QUESTIONS: 5

Total Marks:50

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I G. Srivalli (192365014) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

G. Srivalli

Signature of the student:

TECHNICAL PRESENTATION

1. From Caesar to Vigenère: Evolution of Substitution Ciphers”
2. “Substitution vs Transposition: Which Provides Better Security?”
3. “Security Attacks, Threats, Risks and Their Impact on Security Goals”
4. “Steganography vs Cryptography: Hidden Security in the Digital World”
5. “Role of Number Theory in Cryptography: From Euclid to Euler”

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation



Steganography vs Cryptography

Hidden Security in the Digital World

Two ancient crafts for the same modern problem — keeping digital secrets.

Q3ZKXJAMRWPLF9DUYTGBHSEKOCVZQWIRX7M2

MZWX **H** TJAKR **I** PEYS **D** MFGOB **D** QVCWJ **E** ANBWK **N** ZQ

LPKFHXBVQRITAWZJDMYUCGOSKPLQ8WER5TAZ

the word "HIDDEN" —
hiding in plain sight

Presented by: G. Srivalli
Reg no: 192365014

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01 · THE BIG IDEA

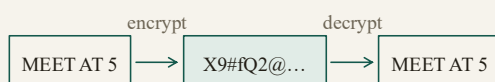
Two Ways to Keep a Secret: Scramble It, or Hide It

CRYPTOGRAPHY

Scramble the message so nobody can read it without the key.

Anyone may see the ciphertext — to them it is useless gibberish.

Like a locked safe in plain sight — everyone sees the safe, only the key opens it.



STEGANOGRAPHY

Hide the message itself inside an ordinary carrier — a photo, a song, a video — so no one even suspects a secret exists.

Like invisible ink in a normal letter — the letter looks routine, the secret stays unseen.

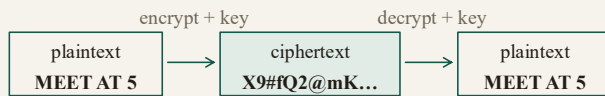


Crypto protects the content. Stego protects the **existence**.

02 · TECHNIQUE ONE

Cryptography: Visible to All, Readable by None

HOW IT WORKS



The gibberish is **meant to be seen** — security lives in the **key**, not in hiding the algorithm (Kerckhoffs's principle).

TWO FLAVORS

Symmetric — **one shared key**. Fast, used for bulk data and storage.

Example: AES.

Asymmetric — **public + private key pair**. Solves secure key exchange.

Example: RSA.

WHERE YOU USE IT DAILY

- The **HTTPS padlock** in your browser
- **End-to-end encrypted** chats — WhatsApp, Signal
- **Chip bank cards** and mobile payments
- Stored passwords, kept as **hashes**

HOW IT IS ATTACKED

Cryptanalysis hunts for mathematical weaknesses;

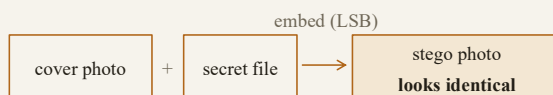
brute force tries every possible key.

With modern key lengths, guessing would take longer than the age of the universe.

03 · TECHNIQUE TWO

Steganography: The Message You Never See

HOW IT WORKS



The **Least Significant Bit** trick: flip the last bit of a pixel's color value. The change is far too small for the eye — but the file now carries a hidden payload.

CARRIERS

Images · audio · video · text — even network packets

A VERY OLD CRAFT

Greek for "*covered writing*" — Herodotus described messages hidden under wax tablets; invisible ink carried the idea into modern warfare.

FRIENDS AND FOES

Legitimate uses

- Digital **watermarking** and copyright tracking
- Protecting journalists' sources and whistleblowers

The dark side

- **Malware** smuggled inside innocent images
- Stolen data exfiltrated past security scanners

HOW IT IS CAUGHT

Steganalysis looks for statistical fingerprints —

unnatural patterns in pixels or sound that betray an embedded payload.

04 · HEAD TO HEAD

Same Goal, Opposite Strategies

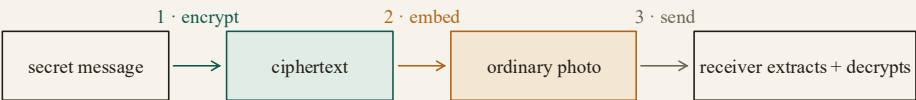
ASPECT	CRYPTOGRAPHY	STEGANOGRAPHY
What it protects	The message's content	The message's existence
What an observer sees	Obvious gibberish — clearly a secret	Nothing suspicious at all
Core method	Scramble with algorithm + key	Embed inside an innocent carrier
Security rests on	Secrecy of the key	Secrecy of the hiding method
Attacked by	Cryptanalysis, brute force	Steganalysis, statistical tests
If defeated	Content leaks once the key cracks	Exposed the moment it is noticed
Everyday example	HTTPS, encrypted messaging	Watermarks, data hidden in media

Crypto assumes the enemy sees the message — stego assumes the enemy must never notice it.

05 · PUTTING IT TOGETHER

The Strongest Secret Is Scrambled and Hidden

LAYERED SECURITY — DEFENSE IN DEPTH



Even if someone discovers the hidden photo, they still face the encryption — **two walls, not one.**

THREE THINGS TO REMEMBER

- 1 **Cryptography** = visible but unreadable. **Steganography** = invisible and ordinary-looking.
- 2 They answer different questions — one guards the content, the other guards the existence.
- 3 Layered together, **finding the message still isn't reading it.**

Thank you — questions?